Speed Regulation of a Flywheel Using Eddy Current Braking

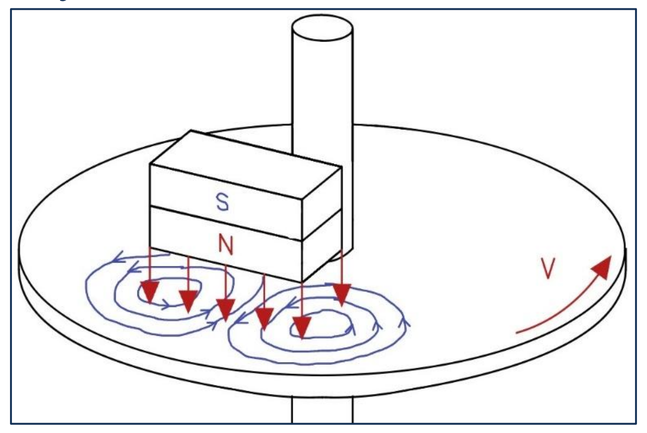
AE 4610 – Group A03-A – Georgia Institute of Technology

Deniz Alpdogan, Garrett Comes, Ella Neal, Zain Pastula, Kaleb Patel, Brent Perdew

Introduction and Motivation

Project Goal: Our aim was to design and implement a closed-loop motor speed control system capable of maintaining a constant rotational speed of about 300 RPM in the presence of externally induced magnetic breaking on a rotating aluminum disk.

Real World Application: The controller design could aid in the development of magnetic braking systems for train and roller coaster wheels. In addition, a similar control system can be developed for wind turbines to allow for a constant rotor speed with varying aerodynamic forces on the external blades.



Magnetic Braking Schematic with Induced Eddy Currents

Controller Design

Controller Type: A Proportional-Integral (PI) controller was selected for this system. The proportional term provides rapid response to changes in speed, while the integral term eliminates steady-state error caused by the nonlinear eddy current braking disturbance. A derivative term was not included due to the relatively smooth system dynamics and to avoid amplifying measurement noise from the encoder. The PI controller offered a good balance between simplicity, robustness, and the ability to maintain the desired constant rotational speed under varying braking conditions.

Attaining Stability: Our team initially utilized the following gains from our python simulator to examine our system's stability. We observed a stable response of our motor and disk upon introducing the magnet with these gain values, but we chose to manually adjust the gains to verify our stability was maximized. Since the system's response was not improved as a result, we chose to use the simulator values as our final gains.

Gain	Value
K_p	0.001
K_i	0.013
K_s	0.350
K_v	0.00213

Desired Performance Criteria:

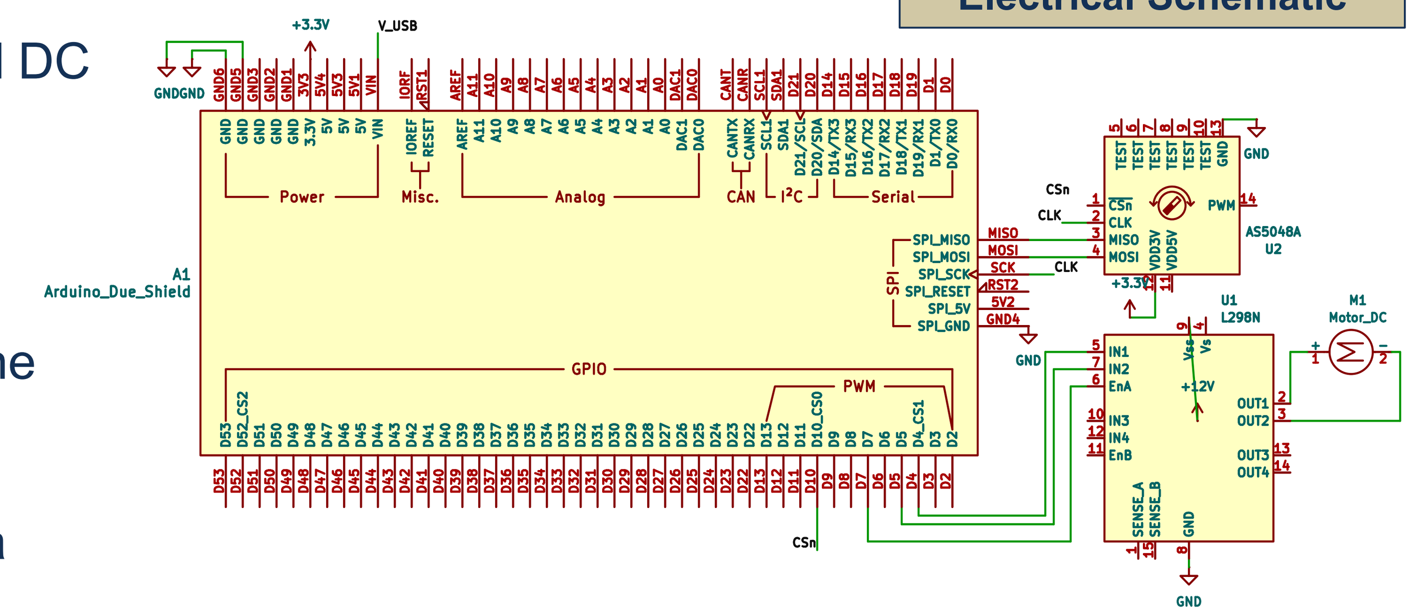
Gain	Value
Target Rotational Speed	300 RPM
Rise Time	3 sec
Settling Time	6 sec
Steady State Error	± 10 RPM
Maximum Overshoot	5%

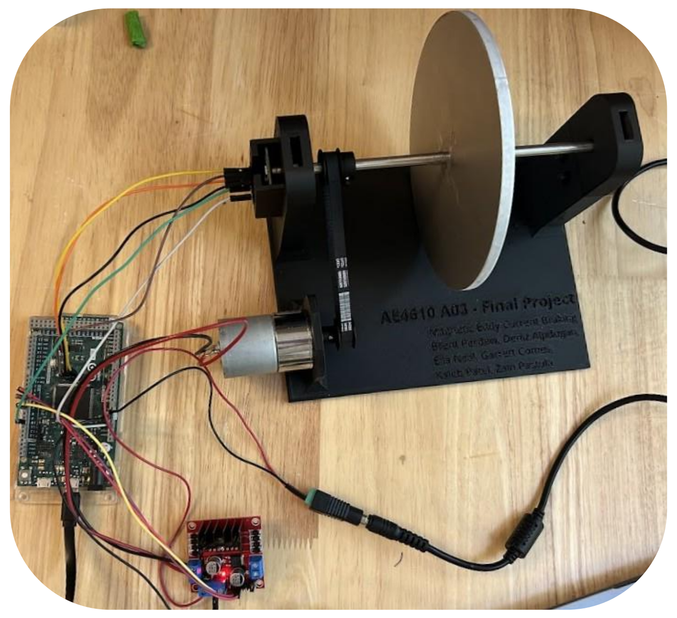
Hardware Setup and System Modeling

System Components:

- **Actuator:** High-torque 550 RPM brushed DC motor driven by PWM controller.
- **Flywheel Assembly:** Precisely balanced aluminum disk on a central shaft.
- **Disturbance Mechanism:** High-strength permanent magnet mounted on stage. The position is manually adjusted to set the magnetic brake force.
- **Control Unit:** Microcontroller executing a robust PID feedback control.

Electrical Schematic





Sensing and Actuation:

Measured Variables & Sensors:

- Rotational Speed (RPM): Optical Encoder (on motor shaft).

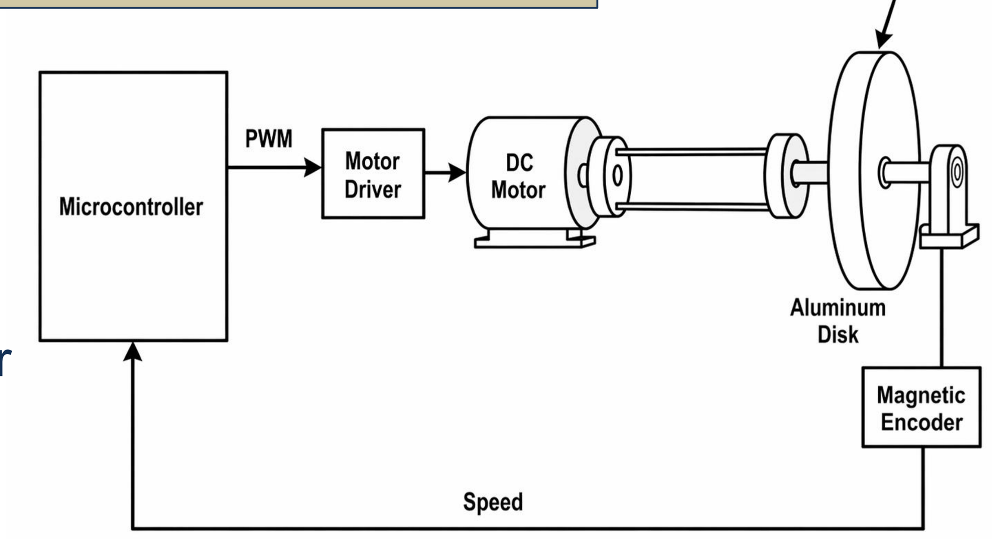
Control Inputs:

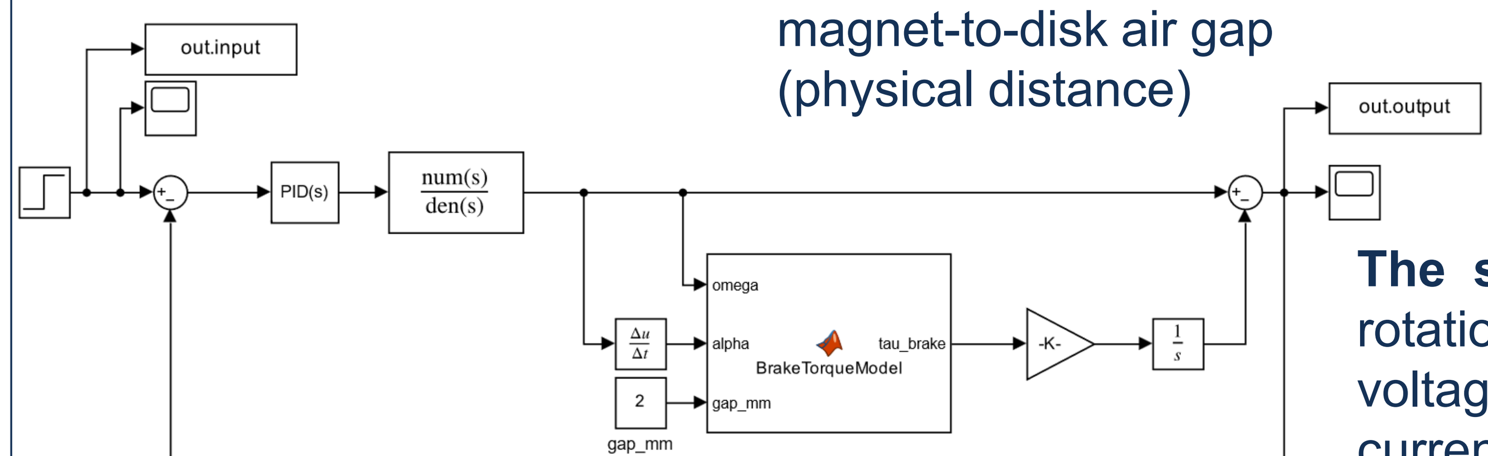
- Motor Voltage: Analog PWM signal command to the motor driver.

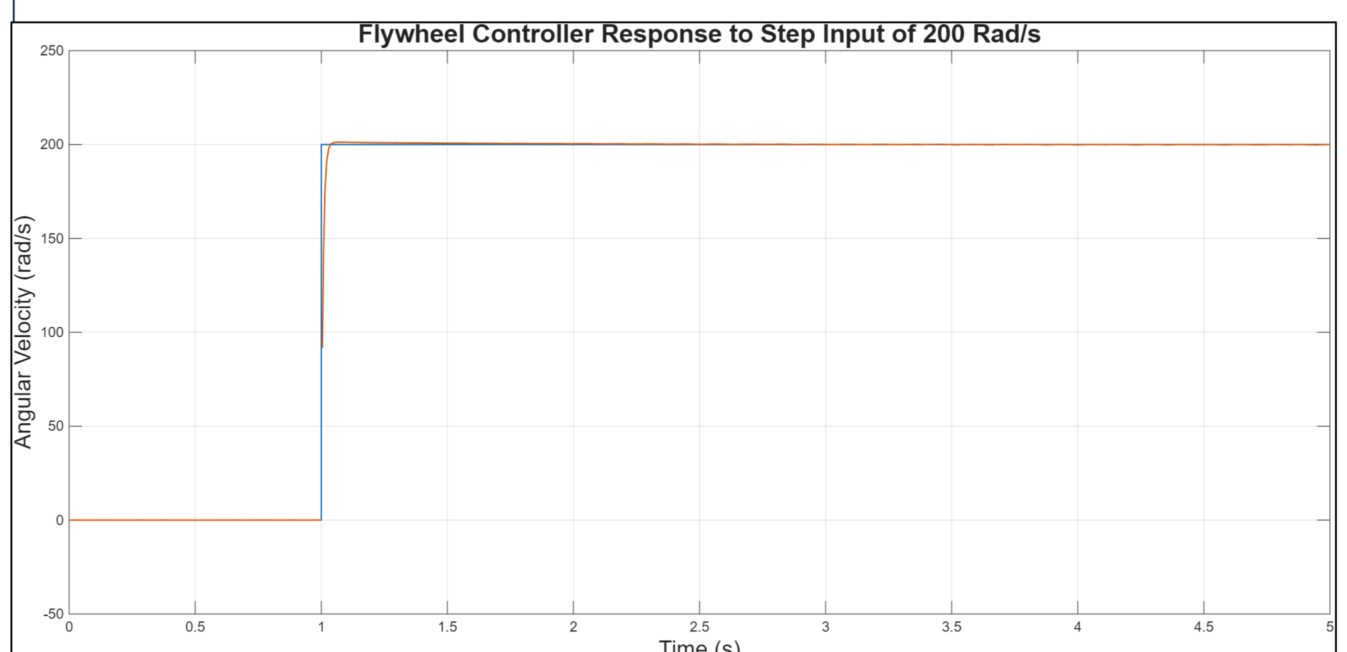
Disturbance Variable:

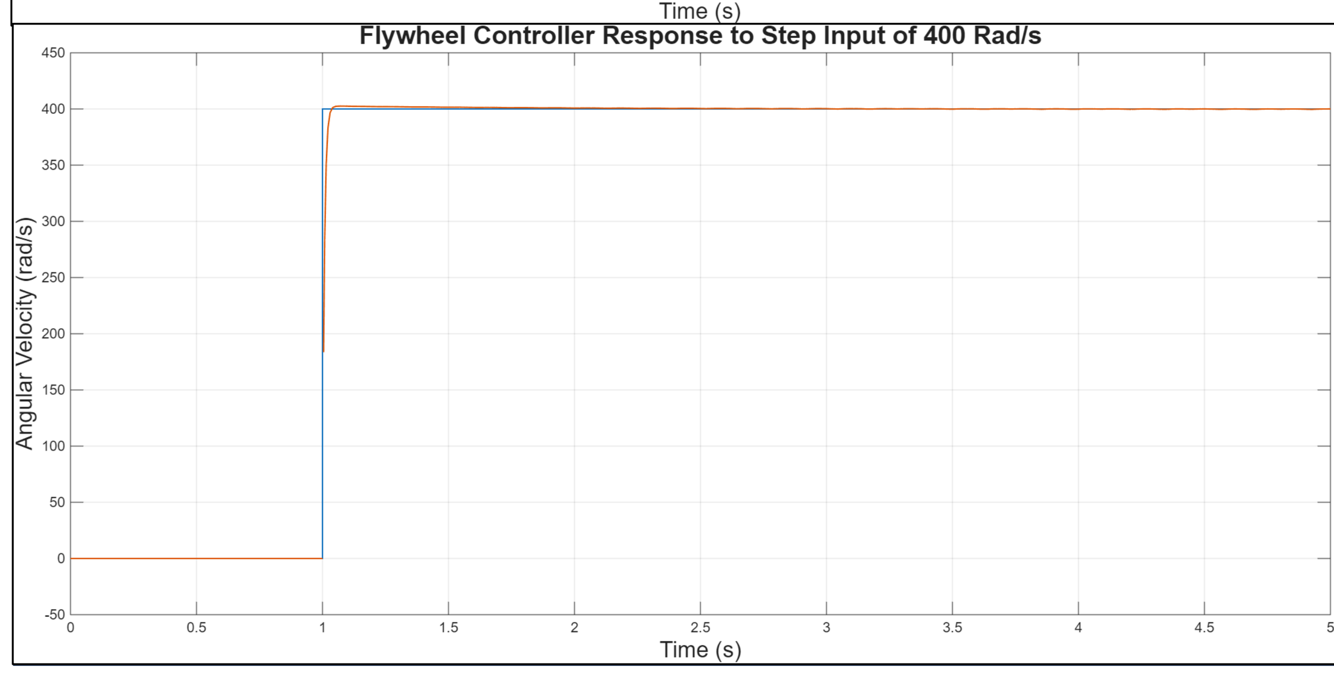
- Magnetic Field Interaction: Manually controlled via the magnet-to-disk air gap (physical distance)

Hardware Diagram





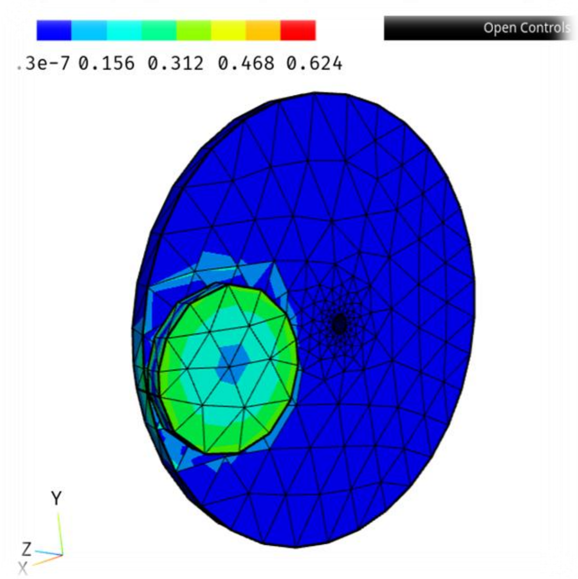




System Modeling

The system was modeled as a closed-loop rotational plant relating motor input (PWM voltage) to flywheel speed. The nonlinear eddy current braking disturbance was incorporated using a Chebyshev polynomial fit, modeling braking torque as a function of angular velocity, angular acceleration, and magnet gap distance.

A Proportional-Integral (PI) controller was implemented to regulate speed, where the proportional term provides fast response and the integral term eliminates steady-state error due to the disturbance. A derivative term was omitted to avoid amplifying encoder noise.



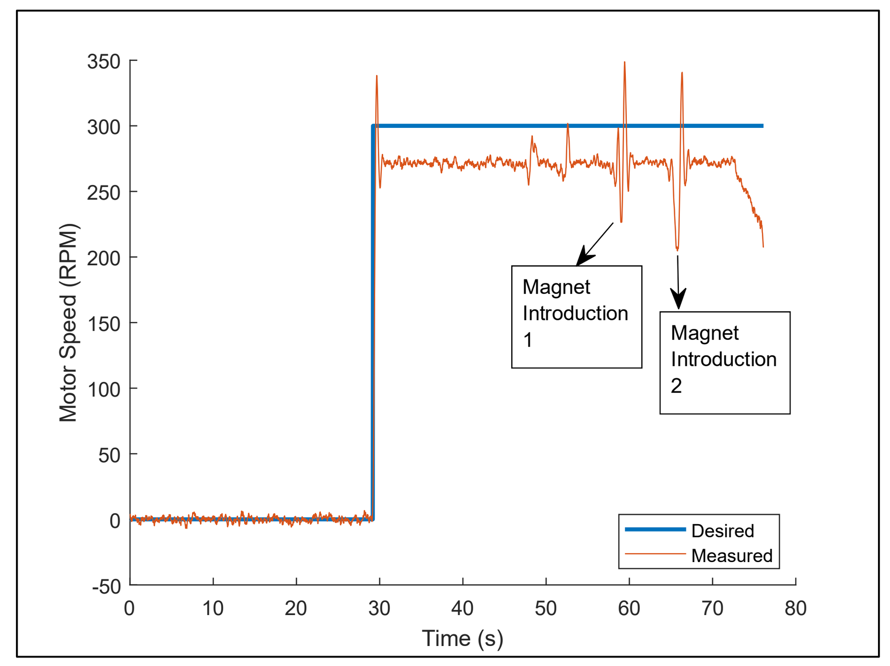
Braking Torque in N-m

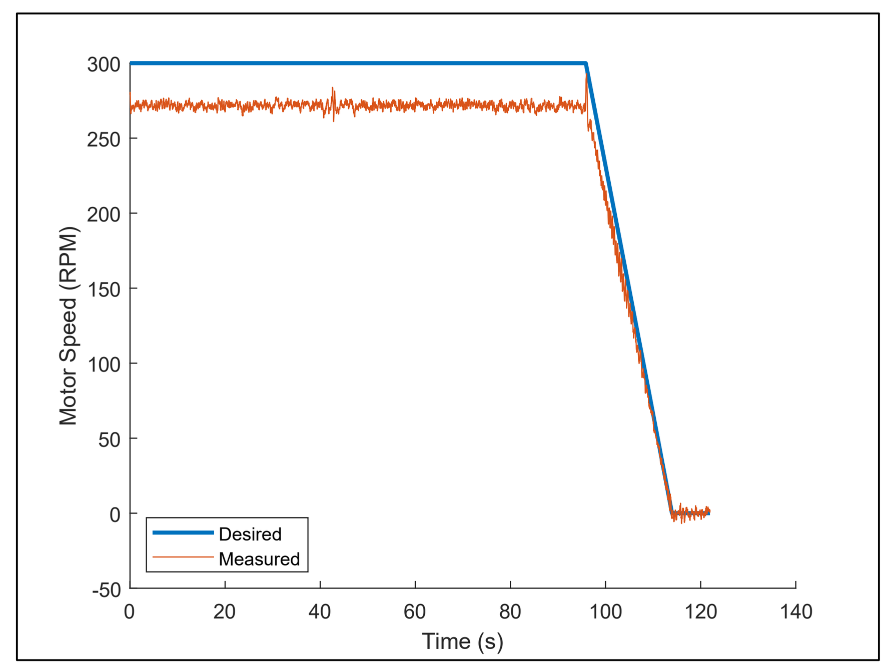
Simulation results show minimal difference between 200 rad/s and 400 rad/s step inputs, as the system behaves approximately linearly over this range and the controller dominates the transient response.

Evaluation, Results, and Conclusion

Conclusion

- **Target Regulation:** The closed-loop PID controller successfully maintained the target rotational speed of 300 RPM despite the non-linear torque load introduced by the magnet
- **Disturbance Rejection:** Upon introduction of eddy current brake, the system demonstrated robust recovery, returning to steady-state within the required 6-second settling time.
- **System Performance:** Initial testing indicates that the rise time and maximum overshoot remain within the design specifications of 3 sec and 5%, respectively.
- **Steady-State Accuracy:** The controller effectively compensated for frictional losses and magnetic drag, keeping the steady-state error within ±10 RPM band.





Future Extensions and Improvements

Disk Concentricity: Current disk exhibits slight eccentricity, which manifests as periodic fluctuations in encoder readings which made the steady state error of ±10 RPM hard to achieve.

Mechanical Power Transmission: Replacing 3D-printed pulleys with metal to improve belt grip; implementing a dedicated pulley tensioner would eliminate belt slip.